

Xindi (Cindy) Wu

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Computer Science Department
Princeton University

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Education

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| 2022–now | Ph.D., Computer Science, Princeton University
Advisor: Olga Russakovsky |
| 2020–2022 | M.S., Computer Vision, Robotics Institute, Carnegie Mellon University
Advisor: Deva Ramanan |
| 2016–2020 | B.S., Computer Science, Honors Youth Program, Xi'an Jiaotong University
Advisors: Jinjun Wang and Pengju Ren |

Research Interests

Computer Vision, World Models, Multimodal Machine Learning.

Award

- **Outstanding Paper Honorable Mention**, ICML 2026, Motion Attribution for Video Generation.
- **Best Paper Award**, ICML 2026 From Frames to Stories Workshop, Addressable Memory for Video World Models.
- **Best Paper Award**, ICLR 2026 Test-Time Update Workshop, Reinforced Fast Weights with Next Sequence Prediction.
- **Apple Scholars in AIML PhD Fellowship**, Apple, April 2026.
- **Best Paper Award**, ECCV 2024 Dataset Distillation Workshop, Vision-Language Dataset Distillation.

Experience

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| • NVIDIA, Spatial Intelligence Lab , Santa Clara, CA | <i>May 2025 – May 2026</i> |
| Research Scientist Intern with Dr. Jonathan Lorraine, Dr. Despoina Paschalidou, Dr. Aljosa Osep, Dr. James Lucas, Dr. Jun Gao, Prof. Laura Leal-Taixé, Prof. Antonio Torralba, Prof. Sanja Fidler | |
| • Meta Reality Lab, Smart Glass AI Team , Redmond, WA | <i>May 2023 – Aug. 2023</i> |
| Research Scientist Intern with Dr. Shane Moon, Dr. Uriel Singer Dr. Luna Dong, Dr. Zhaojiang Lin, Dr. Paul Crook, Dr. Andrea Madotto Dr. Ethan Xu. | |
| • Snap Inc., Perception Team , New York, NY | |
| Machine Learning Engineer | <i>Feb. 2022 – Aug. 2022</i> |
| Research Intern with Dr. Alireza Zareian and Dr. Chen Wang | <i>May 2021 – Aug. 2021</i> |
| • CMU Argo AI Center for Autonomous Vehicle Research , Pittsburgh, PA | <i>Jan. 2021 – Jan. 2022</i> |
| CMU Sponsored Capstone and Research Assistant with Prof. Deva Ramanan. | |

Publications & Preprints

** indicates equal contribution.*

- [30] **Addressable Memory for Video World Models.**
X. Wu, S. Elflein, J. Lucas, O. Russakovsky, L. Leal-Taixé, D. Paschalidou, J. Lorraine, A. Osep.
Preprint 2026. ICML 2026 From Frames to Stories Workshop (Best Paper Award). [Link](#).
- [29] **NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation.**
NVIDIA, A. Basant, ..., X. Wu, ..., Z. Gojcic. *Preprint 2026.* [Link](#).
- [28] **Variance Reduction for Expectations with Diffusion Teachers.**
J. Bettencourt, X. Wu, M. Atzmon, J. Lucas, J. Lorraine. *Preprint, 2026.* [Link](#).
- [27] **Reinforced Fast Weights with Next-Sequence Prediction.**
H. S. Hwang*, X. Wu*, S. Chun, O. Russakovsky.
Preprint 2026, ICLR Test-Time Update Workshop 2026 (Best Paper Award). [Link](#).
- [26] **Motion Attribution for Video Generation.**
X. Wu, D. Paschalidou, J. Gao, A. Torralba, L. Leal-Taixé, O. Russakovsky, S. Fidler, J. Lorraine.
ICML 2026 (Outstanding Paper Honorable Mention, Oral). [Link](#).
- [25] **COMPACT: COMpositional Atomic-to-Complex Visual Capability Tuning.**
X. Wu*, H. S. Hwang*, P. Kirichenko, O. Russakovsky. *ICLR 2026.* [Link](#).
- [24] **Actions as Language: Fine-Tuning VLMs into VLAs Without Catastrophic Forgetting.**
A. J. Hancock, X. Wu, L. Zha, O. Russakovsky, A. Majumdar. *ICLR 2026.* [Link](#).
- [23] **Beyond Objects: Contextual Synthetic Data Generation for Fine-Grained Classification.**
W. Yang, X. Wu, Z. Deng, E. Tureci, O. Russakovsky. *CVPR 2026.* [Link](#).
- [22] **ICONS: Influence Consensus for Vision–Language Data Selection.**
X. Wu, M. Xia, R. Shao, Z. Deng, P. W. Koh, O. Russakovsky. *Preprint 2025.* [Link](#).
- [21] **Explain Before You Answer: A Survey on Compositional Visual Reasoning.**
F. Ke, ..., X. Wu, ..., H. Rezatofghi. *Preprint 2025.* [Link](#).
- [20] **DD-Ranking: Rethinking the Evaluation of Dataset Distillation.**
Z. Li, ..., X. Wu, ..., K. Wang. *Preprint 2025.* [Link](#).
- [19] **Corgi: Cached Memory-Guided Video Generation.**
X. Wu, U. Singer, Z. Lin, A. Madotto, X. Xia, P. A. Crook, Y. E. Xu, X. L. Dong, S. Moon. *WACV 2025.*
[Link](#).
- [18] **ConceptMix: A Compositional Image Generation Benchmark with Controllable Difficulty.**
X. Wu*, D. Yu*, Y. Huang*, O. Russakovsky, S. Arora. *NeurIPS D&B 2024.* [Link](#).
- [17] **Vision-Language Dataset Distillation.**
X. Wu, B. Zhang, Z. Deng, O. Russakovsky. *TMLR 2024. ECCV 2024 Dataset Distillation Workshop (Best Paper Award)* [Link](#).
- [16] **SWE-bench Multimodal: Do AI Systems Generalize to Visual Software Domains?**
J. Yang*, C. Jimenez*, ..., X. Wu, ..., O. Press. *ICLR 2025.* [Link](#).

- [15] **CharXiv: Charting Gaps in Realistic Chart Understanding in Multimodal LLMs.**
Z. Wang, ..., **X. Wu**, ..., D. Chen. *NeurIPS D&B* 2024. [Link](#).
- [14] **Language Models as Science Tutors.**
A. Chevalier, ..., **X. Wu**, ..., D. Chen. *ICML* 2024. [Link](#).
- [13] **Pix2Map: Cross-modal Retrieval for Inferring Street Maps from Images.**
X. Wu, K. Lau, F. Ferroni, A. Osep, D. Ramanan. *CVPR* 2023. [Link](#).
- [12] **Ego4D: Around the World in 3,000 Hours of Egocentric Video.**
K. Grauman, ..., **X. Wu**, ..., J. Malik. *CVPR* 2022 (*Best Paper Finalist, Oral*). [Link](#).
- [11] **Toward Learning Robust and Invariant Representations with Alignment Regularization and Data Augmentation.**
H. Wang, Z. Huang, **X. Wu**, E. P. Xing. *KDD* 2022. [Link](#).
- [10] **CryoETGAN: Cryo-electron Tomography Image Synthesis Using Unpaired Image Translation.**
X. Wu, C. Li, H. Wei, H. Deng, J. Zhang, M. Xu. *Frontiers in Physiology* 2022. [Link](#).
- [9] **Squared ℓ_2 Norm as Consistency Loss for Leveraging Augmented Data to Learn Robust and Invariant Representations.**
H. Wang, Z. Huang, **X. Wu**, E. P. Xing. *arXiv* 2021. [Link](#).
- [8] **Marrying Motion Forecasting and Offline Model-Based Reinforcement Learning for Self-Driving Cars.**
S. Pande, **X. Wu**. *Preprint* 2021. [Link](#).
- [7] **High Frequency Component Helps Explain the Generalization of Convolutional Neural Networks.**
H. Wang, **X. Wu**, Z. Huang, E. P. Xing. *CVPR* 2020 (*Oral*). [Link](#).
- [6] **Transferable Adversarial Attacks on Deep Reinforcement Learning.**
X. Pan, Y. Cao, **X. Wu**, E. Zelikman, C. Xiao, Y. Sui, R. Chakraborty, R. S. Fearing. *CVPR 2020 Workshop on Adversarial ML*. [Link](#).
- [5] **Reducing Exploitation of Data Idiosyncrasy Helps Robustify Trained Models.**
X. Wu, H. Wang, E. Zelikman, M. Xu, E. P. Xing. *Preprint* 2020. [Link](#).
- [4] **Regularized Adversarial Training (RAT) for Robust Cellular Electron Cryo Tomograms Classification.**
X. Wu, Y. Mao, H. Wang, X. Zeng, X. Gao, E. P. Xing, M. Xu. *BIBM* 2019 (*Oral*). [Link](#).
- [3] **Template-based and Template-free Approaches in Cellular Cryo-electron Tomography Structural Pattern Mining.**
X. Wu, X. Zeng, Z. Zhu, X. Gao, M. Xu. *Computational Biology (Codon Publications)* 2019. [Link](#).
- [2] **Deep Self-Paced Learning for Semi-supervised Person Re-identification Using Multi-View Self-Paced Clustering.**
X. Xin, **X. Wu**, Y. Wang, J. Wang. *ICIP* 2019. [Link](#).
- [1] **Multitask Learning with Enhanced Modules.**
Z. Zheng, Y. Wei, Z. Zhao, **X. Wu**, Z. Li, P. Ren. *DSP* 2018. [Link](#).

Talks & Poster Presentations

- **Addressable Memory for Video World Models.** ICML From Frames to Stories Workshop oral presentation, Seoul, Jul. 2026.
- **Motion Attribution for Video Generation.** ICML oral presentation, Seoul, Jul. 2026.
- **Tracing Influence in Video World Models.** NVIDIA, May 2026.
- **Visual Compositional Tuning.** ICLR poster presentation, Rio de Janeiro, Brazil, Apr. 2026.
- **Scaling Multimodality through a Data-Centric Lens.** NYU, Apr. 2026.
- **Where is Motion From? Scalable Motion Attribution for Video Generation Models.** ICCV Reliable and Interactive World Model Workshop, Honolulu, Oct. 2025.
- **From Data to Capability: Data for efficient multimodal machine learning.** Datology AI Summer of Data Seminar, June 2025.
- **ConceptMix: A Compositional Image Generation Benchmark with Controllable Difficulty.** NeurIPS, Vancouver, Dec. 2024.
- **Vision-Language Dataset Distillation.** ECCV Dataset Distillation Workshop, Milan, Oct. 2024.
- **ConceptMix: A Compositional Image Generation Benchmark with Controllable Difficulty.** ECCV Knowledge in Generative Models Workshop, Milan, Oct. 2024.
- **Compositional Generation Evaluation.** Google Research, New York, July 2024.
- **Scaling Down before Scaling Up: Recent Progress on Dataset Distillation.** CVPRW Dataset Distillation, Seattle, June 2024.
- **Corgi: Compositional Memory-Guided Video Generation.** NYC Vision Day, New York, Nov. 2023.
- **Pix2Map: Cross-modal Retrieval for Inferring Street Maps from Images.** CVPR, June 2023.
- **Regularized Adversarial Training for Robust Cellular Electron Cryo Tomograms Classification.** BIBM, San Diego, Nov. 2019.

Professional Service

- **Co-organizer:** Continual World Models (NeurIPS 2026); Curated Data for Efficient Learning Workshop (ECCV 2026/ICCV 2025)
- **Reviewer:** NeurIPS 2025/2024/2023; ICLR 2026/2025/2024; CVPR 2026/2025/2024/2023/2022; ICML 2025/2024; ICCV 2025/2023; ECCV 2024/2022; TMLR; ICRA 2024; ACCV 2024; ICLR 2023 Workshop ME-FoMo; NeurIPS Interpolate Workshop 2022; BMVC 2020; IJCAI 2020.
- **Committee Member:** Diversity, Equity and Inclusion Committee, Robotics Institute, CMU.
- **Volunteer:** ECCV 2024, vGHC (Grace Hopper Celebration) 2021.
- **Panelist:** Robotics Institute MS Student Panel 2021; Robotics Institute Summer Scholars (RISS) 2021.

Teaching

- **Teaching Assistant:** COS 429 Computer Vision (Princeton), Spring 2024. Instructors: Vikram V. Ramaswamy and Felix Heide.
- **Teaching Assistant:** COS 597O Advanced Topics in Computer Science: Deep Generative Models (Princeton), Fall 2023. Instructor: Adj. Bousso Dieng.

Last updated: July, 2026.